INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR

DEPARTMENT OF PHYSICS

MID-AUTUMN SEMESTER EXAMINATION 2024-25

Subject: **Classical Mechanics (PHPH31207)** Total Marks: **30**

Time Duration: **3 Hours**

No. of students: **99** Date: **23-09-2024** & Time: **2 PM- 4 PM**

Answer all the questions (Start each question on a fresh page)
(If the given information is not sufficient make proper assumptions and mention them clearly)
(Take g = 10 m/s^2, e = 2.72 where ever it is necessary)

1. A particle of mass 0.01 kg falls freely in the earth's gravitational field with an initial velocity 15 m/s. If the air exerts a frictional force of the form f = -kv, where k = 0.05 N-s/m and v is velocity then find the velocity of the particle at time 0.2 s. (Use Newtonian Mechanics) (5M)
2. (a) Prove that the shortest distance between two points in space is a straight line. (2M+3M)
   (b) Obtain Newton's second law from Hamilton's principle.
3. Deduce Lagrange's equation using Hamilton's principle for non-conservative and holonomic system. (5M)
4. Obtain the Lagrangian and Hamiltonian of a Charged particle in an electro magnetic field. (5M)
5. A mass M sliding freely along a friction less rail. A pendulum of length l and mass m hangs from M as shown in the Fig. 1. Find the equation of motion of the pendulum. (5M)
6. Find the Lagrangian of a system of mass m whose Hamiltonian is given as $H(p,r) = \frac{p^2}{2m} - (\vec{a} \cdot \vec{p})$, where $\vec{a}$ = constant and $\vec{p}$ is momentum. (5M)

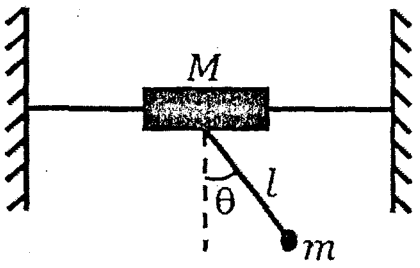

Fig. 1

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